Crypto Cash-and-Carry: Capital Efficiency in the Spot–Perpetual Basis Trade

Summary version. Full paper: `docs/paper.pdf`.

Question

Buying spot crypto and shorting the perpetual future ("perp") collects the perp's **funding rate** with no price exposure: economically, a secured loan of cash. Funding has an exchange-set floor of 0.01% per 8 hours (**10.95% a year**). Can the trade be improved by forecasting funding, or by optimising which coins to hold and how much?

> **Excess return over T-bills ≈ funding ÷ margin factor − T-bill rate − trading costs**, where the margin factor is the capital tied up per $1 of notional.

Approach

- **Data:** Binance and Bybit, 2021–2026: funding, volume and 8-hour prices for 274 coins, including later delisted ones. Each exchange is a separate $500k book.
- **Forecasts:** four models of each coin's next-7-day funding, trained on 2021 and tested out of sample from 2022 with monthly walk-forward refits.
- **Strategies:** passive BTC+ETH; rotating into the top 5 by forecast, with and without a cost hurdle; and a daily mixed-integer optimiser (MILP, HiGHS) choosing coins and sizes under capital, liquidity and per-coin limits.
- **Discipline:** every decision rule written down before the test ran; every variant (23) counted when judging significance.

Findings

- **Carry over cash collapsed after 2021.** Passive BTC+ETH earned +12.6% over T-bills in 2021, then +0.4% a year in 2022–26: the funding floor stayed fixed while T-bills rose from 0% to 5%.
- **Funding is forecastable, but the edge is the size of the cost.** A pooled regression ranks coins well (rank IC about 0.5): the top fifth earns 9.7% a year more than the average coin. But about half of it changes each week, and rotating costs 10.4% a year.
- **No active strategy beat passive** at the same margin on a risk-adjusted basis, and none survives the correction for the number of variants tried. Naive rotation lost to costs; the cost-aware rule sat mostly in cash; the MILP gave its carry back in about 90 opens and closes a year; small coins brought liquidations.
- **Capital structure decides the outcome.** The trade breaks even at about **1.4×** capital per $1 of notional. Under portfolio margin (1.05×), passive BTC+ETH earned **+1.9% a year over T-bills (Sharpe 4.2)**, the only result that survives the multiple-testing correction.

2022–2026	Margin	Excess return a year	Sharpe	Turnover (× capital)
Passive BTC+ETH	1.50	+0.36%	1.10	0.1
Naive top 5	1.50	−4.31%	−1.68	31.3
Cost-aware top 5	1.50	+1.46%	0.83	3.0
MILP	1.50	+0.63%	0.55	5.2
MILP with turnover controls	1.05 / 1.30	+1.45%	1.21	4.2
Passive BTC+ETH, portfolio margin	**1.05**	**+1.94%**	**4.18**	0.2

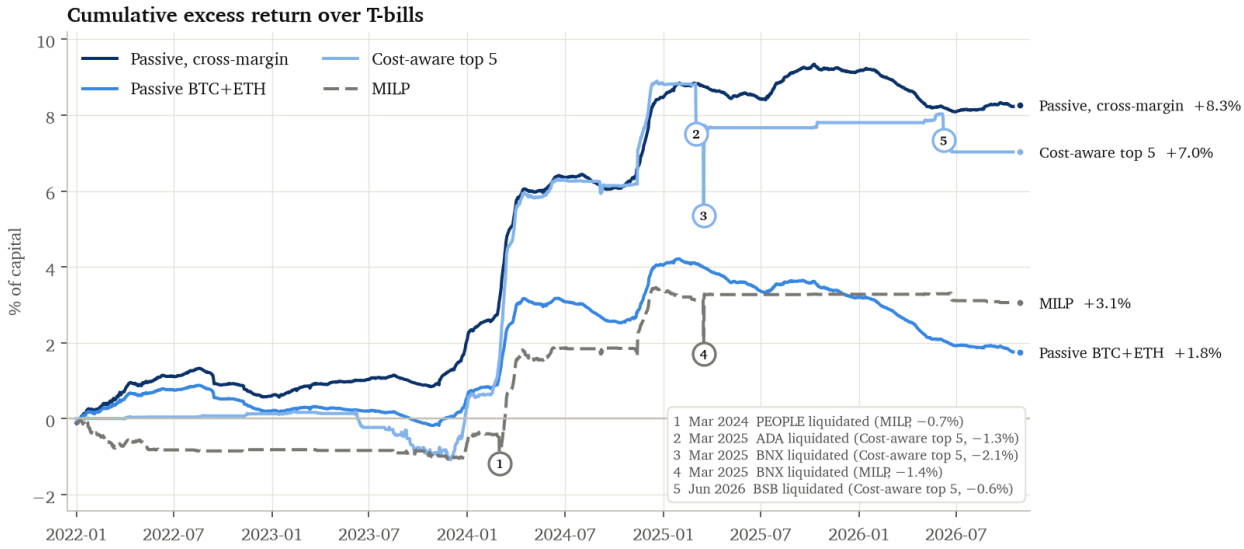


Figure 1. Cumulative return over T-bills, $1M across both exchanges, after all costs (2022–2026). Numbered dips are short-perp liquidations on small coins.

Conclusion

The carry that survived 2021 is a **financing spread**: funding near an exchange-set floor, earned against the cost of cash. Selection and optimisation cannot improve it after costs. What moves returns is the balance sheet: **the margin factor, whether collateral earns interest, and execution cost**.

Limitations: five out-of-sample years with most of the P&L in 2024; low-volatility carry flatters the Sharpe ratio (exchange failure and de-pegs barely appear in the data); 8-hour prices, flat base-tier fees and a simplified liquidation and margin model.